MSE-1 / ROBOTIC AGENTIC AUTOMATION

Intelligent Placement System

From an unstructured job description to an explainable candidate shortlist.

INGEST
JD PDF

INTERPRET
Gemini + Pydantic

MATCH
Student shortlist

AI-ASSISTED SCREENING | PDF + EXCEL | RAG + AGENTS

Evaluation presentation

Why placement screening needs intelligence

- Job descriptions are semi-structured documents: role, responsibilities, mandatory skills, and optional skills are mixed together.
- Student profiles live in a separate Excel workbook, so every placement drive requires repeated manual comparison.
- Manual screening becomes slow, inconsistent, and difficult to audit as the number of candidates grows.
- The target outcome is a shortlist with a visible path from job requirement to candidate decision.

THE INPUT REALITY

Job Description PDF document
Student Database 100 Excel records

THE DECI

Who is elig
matched? V
next?

SUCCESS MEASURE

**Reduce repetitive screening while pre
traceability from JD evidence to cand**

A focused RAG and agent workflow

STUDENT DATA

Excel workbook

Pandas DataFrame / 100 rows

RETRIEVE

ELIGIBILITY AGENT

Skill matching

Normalize -> compare -> shortlist

OUTPUT / EXPLAINABLE CANDIDATE SHORTLIST

What makes the prototype intelligent

1 / RETRIEVAL

The system retrieves the most relevant JD context before asking the model to interpret it. This keeps analysis grounded in the source document.

2 / STRUCTURE

Gemini returns a validated JDRequirements object instead of loose text. Downstream code receives predictable fields.

3 / MATCHING

A separate eligibility agent ap
every student skill list and ret
for review.

STRUCTURED REQUIREMENTS

job_role	qualification	10th / 12th / graduation	technical_skills	experience

IMPORTANT TRANSPARENCY

The current prototype extracts academic thresholds but applies technical-skill matching only. Academic filtering, ranking, a reasons are the next release improvements.

Evidence, feasibility, and the next step

DATA EVIDENCE / 100 STUDENT RECORDS

FEASIBILITY

Python-based Local PDF + Excel inputs Batch-ready pipeline Incremental architecture

NOVELTY

RAG-groun output Age matching

THE TAKEAWAY

The system converts an unstructured job description and a student Excel database i repeatable, AI-assisted, and explainable screening workflow.

Recommended next demo: show the JD extraction, run the shortlist, and explain one candidate decision end-to-end.